Emotion Detection in Children's Drawings: Predicting Emotional Trends and At-Risk Students with Deep Learning

Emotion Detection in Children's Drawings

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1. ABSTRACT

Children's drawings offer critical nonverbal insights into their emotional states. Current approaches largely focus on facial expressions and textual analysis, neglecting the symbolic nature of children's art. This project introduces an interpretable machine learning model in free-hand children's drawings to classify basic emotions—happy, sad, angry, fearful. Using a dataset of 1,109 labeled images, we extracted visual features such as edges, shapes, and color statistics, and trained a MobileNetV2-based classifier, achieving an overall F1-score of 0.63. Additionally, we aggregated predictions per child using a Random Forest model, achieving 100% accuracy in identifying at-risk students on a small test set that we created.

2. INTRODUCTION

Early diagnosis of emotional and developmental disorders in children is challenging due to communication barriers between children and adults. Psychologists often interpret children's drawings as a method to assess emotional health, yet manual evaluation is subjective and inconsistent, potentially causing misdiagnosis. Computational methods promise objectivity and scalability, but research in analyzing children's artwork remains limited. This paper addresses this gap by developing a robust, automated system capable of accurately classifying emotions from unstructured visual data.

3. RELATED WORKS

Current research in emotion recognition has primarily targeted facial expressions and text. For instance, in facial expression analysis, Savchenko (ICML 2023) proposed a frame selection approach to reduce computational overload through selectively processing video frames. In text recognition, Wen et al. (TACL 2023) introduced refined emotional labels in textual conversations to better capture the mixed emotions in dialogues. Additionally, Zhang et al. (EMNLP 2024) explored visual prompting in Large Language Models (VLLMs) to improve emotional cue localization across different modalities.

While research in these fields is particularly useful in the advancement of automating human affect understanding, relatively little work has been done involving emotion detection in artwork, more specifically, children's drawings. This project uniquely focuses on the symbolic, abstract nature of children's drawings, extending computational emotion recognition research into a new domain.

4. METHODOLOGY

4.1 DATA PREPROCESSING

We applied the following preprocessing steps to prepare the dataset for training:

- Converted images to JPEG format - to ensure consistency in file formatting and compatibility with TensorFlow image pipeline

- Normalized pixel values [0, 1] - to improve convergence during training by stabilizing the learning process
- Resized images to 224x224 pixels - to match the input size expected by the models used in training
- One-hot encoded emotion labels - converts the categorical emotion classes into a machine-readable format to perform multiclass classification
- Created TensorFlow dataset with batching (32), shuffling, and prefetching - used to optimize loading efficiency and improve training speed through overlapping data preprocessing and model execution
- Split the dataset into training (80%) and testing (20%) sets - used to evaluate the model on unseen data
- Further split the training set into training and validation sets with an 80-20 ratio - used to monitor overfitting and tune hyperparameters during model development

4.2 MODEL ARCHITECTURE

We built a fine-tuned Convolutional Neural Network (CNN):

- **Base Model**: MobileNetV2 pretrained on ImageNet, with the first 70 layers frozen - freezing early preserves generic visual features, allowing for reduction in training time and overfitting prevention from our relatively small dataset
- **Feature Extraction**: Global average pooling - reduces spatial dimension of feature maps to a single vector per channel, lowering parameter count and providing spatial invariance
- **Classification Head**:
    - Dense(512, ReLU) - adds capacity to learn high-level representations specific to children's drawing features
    - Dropout(0.35) - mitigates overfitting by randomly dropping neurons during training
    - Dense(256, ReLU) - further refines discriminative features before performing the final classification
    - Dropout(0.35)
- **Output Layer**:
    - Dense(4 units, softmax) for the 4 emotion classes: provides normalized class probabilities for multiclass emotion classification

4.3 TRAINING CONFIGURATION

- **Optimizer**: Adam - Adam optimizer adapts learning rates for each parameter, working well in practice
- Learning rate = 0.00001 - low learning rate to fine-tune the pretrained model
- **Loss**: Categorical cross-entropy - used for multiclass classification tasks where one-hot encoded labels are used
- **Batch size**: 32 - balances memory use and training stability
- **Epochs**: 100 - allows enough iterations for convergence
- **Early stopping**: Patience of 10 epochs - halts training when validation loss doesn't improve for the next 10 consecutive epochs to prevent overfitting

4.4 EVALUATION METRICS

- **Overall accuracy** - measures general classification performance
- **Classification Report** (precision, recall, and F1-score per emotion class) - evaluation of classwise performance and balance between false positives and false negatives
- **Confusion matrix** - insights into common misclassifications and error patterns

4.5 TOOLS AND FRAMEWORKS

- **Python**
- **TensorFlow** - for model building and training
- **Scikit-learn** - for metrics calculation and preprocessing utilities
- **OpenCV** - for image processing tasks
- **Matplotlib** - to visualize the results and training process

5. Experiments and Results

5.1 Experiment Setup

The dataset used in this study consists of 1,109 emotion-labeled children's drawings. We split the dataset into training and testing sets in an 80/20 ratio, before further splitting the training set into training and validation with an 80-20 split, resulting in 710 images for training, 177 images for validation, and 222 for testing. To ensure that each subset maintained the same class distribution, we employed stratified sampling based on the encoded emotion labels.

To improve model generalization and reduce overfitting, we implemented data augmentation using TensorFlow's Sequential API. The augmentations applied included random horizontal flips, ±30% rotations, ±20% zoom, ±30% contrast adjustment, and ±10% translations in both horizontal and vertical directions. These transformations helped the model better learn from limited and abstract visual patterns.

Given the class imbalance in our dataset, we computed balanced class weights using sklearn.utils.class_weight and incorporated them during training to reduce the impact of overrepresented or underrepresented classes. The training process also included early stopping with a patience of 10 epochs, as well as learning rate reduction on plateau—halving the learning rate if validation loss did not improve over 3 consecutive epochs.

For evaluation, we used macro-averaged F1-score and confusion matrix analysis. Before computing the metrics, the one-hot encoded labels were converted back to integer indices to match the model predictions.

5.2 Baseline Comparison

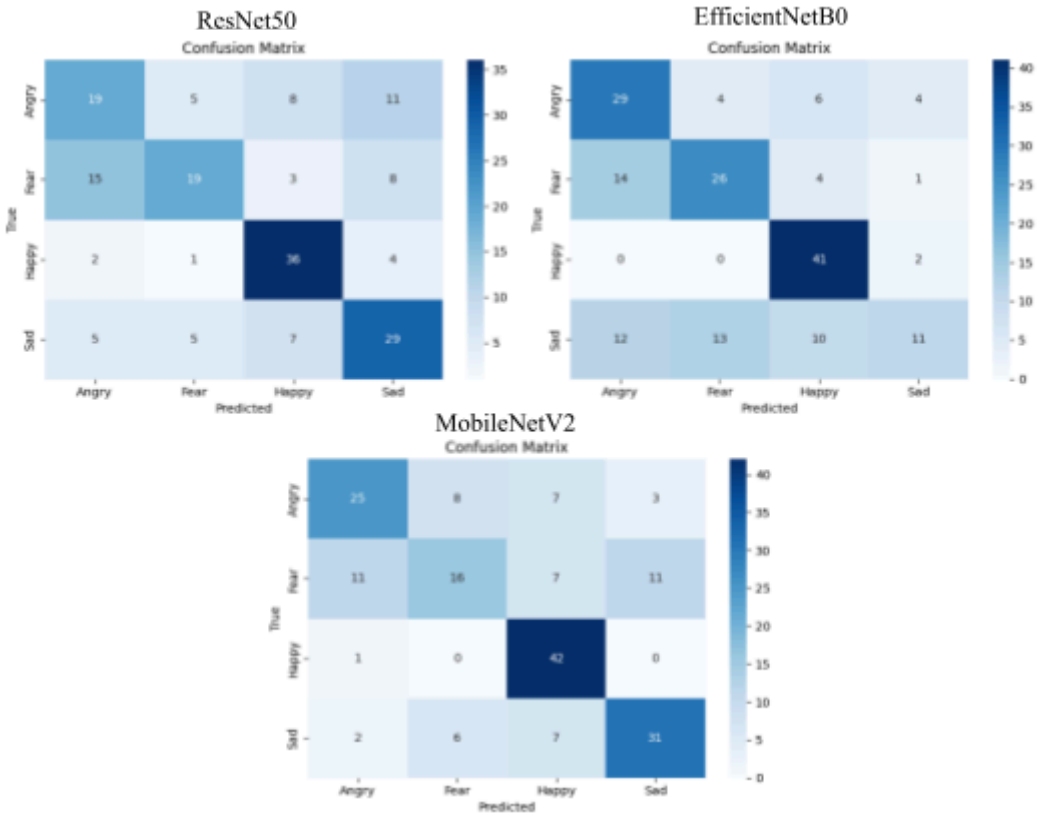


Figure 1: Confusion matrix for the ResNet50, EfficientNetB0, and MobileNetV2 models (top left, top right, bottom middle, respectively)

Model	Accuracy	F1-Score
ResNet50	0.58	0.5733
EfficientNetB0	0.6	0.5787
MobileNetV2	0.64	0.6275

Table 1: Model comparison for the ResNet50, EfficientNetB0, and MobileNetV2

Initially, we planned on training a ResNet50 on our data, but after tuning the model several times with not much improvement in both accuracy and F1-scores, we decided to switch the model to the EfficientNetB0, which was meant to be better for smaller data sets. After re-tuning the hyperparameters once again, we didn't notice a significant improvement in accuracy and F1-scores with failure to reach our accuracy and F1-score goals of 0.6. We did further research and decided to switch the model once again, as we felt that the small data size was influencing the model's performance. Finally, we tested the MobileNetV2 model, and we were able to achieve much higher accuracy and F1-score, so we used it as our base model for our emotion classification task.

As shown in the table:

ResNet50 achieved an accuracy of 0.58 and an F1-score of 0.5733 in our experiments. Although the ResNet family is widely used for image classification tasks, its performance in this task was relatively modest. This may be due to its architectural design being better suited for large-scale or detail-rich image datasets, demonstrating that its poor effectiveness is limited when applied to our relatively small dataset, consisting of abstract and symbolic children's drawings. The confusion matrix for ResNet50 further illustrates this limitation, showing substantial misclassification between negative emotions such as *Fear* and *Anger*, and relatively weak distinction across categories overall.

EfficientNetB0 performed slightly better than ResNet50, with an accuracy of 0.60 and an F1-score of 0.5787. The EfficientNet series is known for its lightweight architecture and efficient parameter scaling, often yielding good performance on moderately sized datasets. In our case, EfficientNetB0 showed improved separation for the *Happy* class, as seen in the confusion matrix, but still struggled with accurately detecting *Fear* and *Sad*, indicating difficulty in handling subtle or overlapping emotional features within drawings, or potentially being issues related to poor classification of smaller data sets, similar to the issue that we found when training the ResNet50 model.

MobileNetV2 significantly outperformed the other two models, achieving the highest accuracy of 0.64 and an F1-score of 0.6275. As a lightweight architecture designed for efficient feature extraction, MobileNetV2 is especially well-suited for processing low-resolution, abstract images. Its streamlined structural design and optimized layer configuration allowed it to better capture the visual and symbolic patterns present in children's drawings, leading to superior performance in our task. The corresponding confusion matrix shows a clearer diagonal dominance, especially for *Happy* and *Sad*, reflecting stronger classification consistency. Although *Fear* remains the most commonly misclassified emotion, MobileNetV2 reduced this confusion compared to the other models.

In summary, MobileNetV2 achieved better classification results with fewer parameters and greater computational efficiency, confirming its suitability for abstract and symbolic visual tasks. Considering both performance and misclassification behavior shown in the confusion matrices, we selected MobileNetV2 as our final model for the subsequent experimental analysis.

5.3 Main Model Performance

Emotion	Precision	Recall	F1-Score
Angry	0.64	0.58	0.61
Fear	0.53	0.36	0.43

Happy	0.67	0.98	0.79
Sad	0.69	0.67	0.68

Table 2: Classification report for the MobileNetV2 model.

The final MobileNetV2 model achieved an overall **F1-score of 0.6275**, demonstrating reasonable effectiveness in classifying emotional states from children's drawings. Performance varied across emotion categories. The model performed best on *Happy* and *Sad*, with F1-scores of 0.79 and 0.68, respectively, indicating that these emotions were easier to detect, possibly due to more distinctive visual patterns or clearer symbolic cues. *Angry* achieved a moderate F1-score of 0.61, while *Fear* proved the most challenging with an F1-score of 0.43, likely due to subtle or overlapping features with other negative emotions. These results suggest that while the model is effective overall, improvements are needed, particularly in distinguishing fear-related expressions, possibly through enhanced feature engineering or additional training data.

5.4 VISUALIZATION

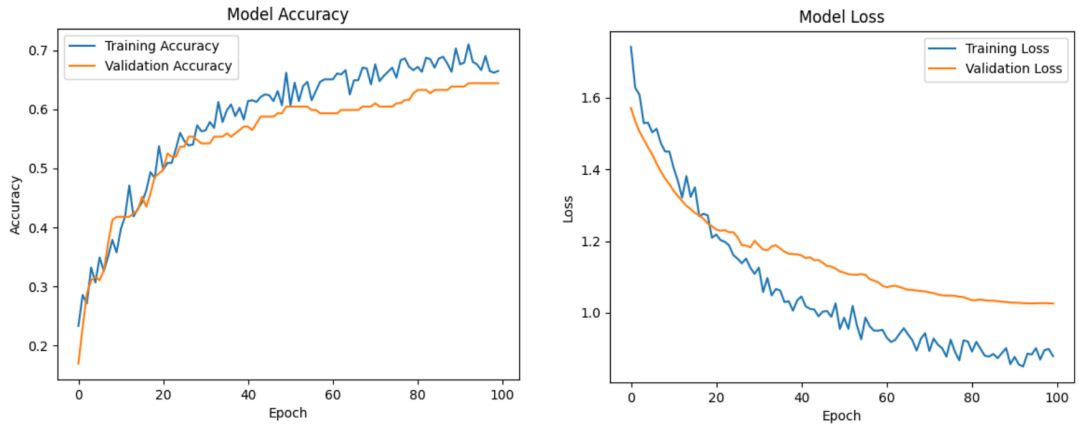


Figure 2: Model accuracy and model loss plots for the MobileNetV2 Model.

The **confusion matrix** confirms class-specific performance differences observed in the F1-scores. The *Happy* class exhibits the clearest separation, with 42 out of 43 samples correctly classified. *Sad* also shows strong performance, though some confusion remains with *Fear* and *Happy*. *Fear* demonstrates the highest misclassification rate, with significant overlap with *Angry* and *Sad*, supporting its lower F1-score. This suggests that visually ambiguous emotional cues and class imbalance are the primary sources of error.

The **accuracy curve** shows steady improvement in both training and validation accuracy, with the validation accuracy plateauing around 0.64. This suggests the model is generalizing reasonably well to unseen data, without signs of overfitting.

The **loss curve** displays a consistent decrease for both training and validation sets. The validation loss stabilizes after around 80 epochs, indicating convergence of the model. The gap between training and validation loss is relatively small, further confirming that the model has learned a stable representation without severe overfitting.

5.5 TESTING SET EVALUATION

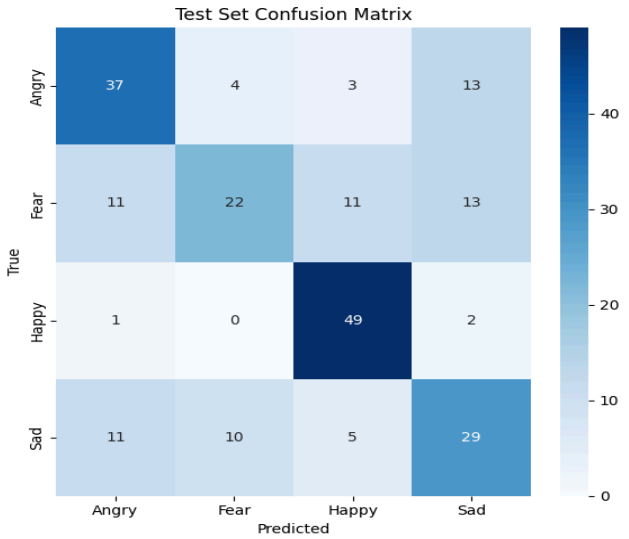


Figure 3: Confusion matrix for test set performance across four emotion classes (Angry, Fear, Happy, Sad).

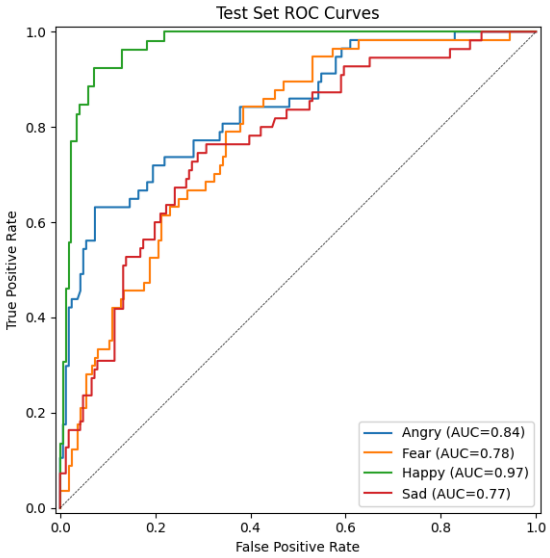


Figure 4: ROC curves for the test set using one-vs-rest classification.

To evaluate the generalization performance of the emotion classification model, we assessed its predictive accuracy on an independent test set. The confusion matrix (Figure 3) revealed strong classification performance, especially for the "Happy" category, though some misclassifications occurred among negative emotions ("Angry," "Fear," and "Sad"). ROC curves (Figure 4) indicated excellent discriminatory ability for "Happy" (AUC=0.97), with moderate performance for the other three emotions (AUC between 0.77–0.84).

5.6 Risk Detection Results

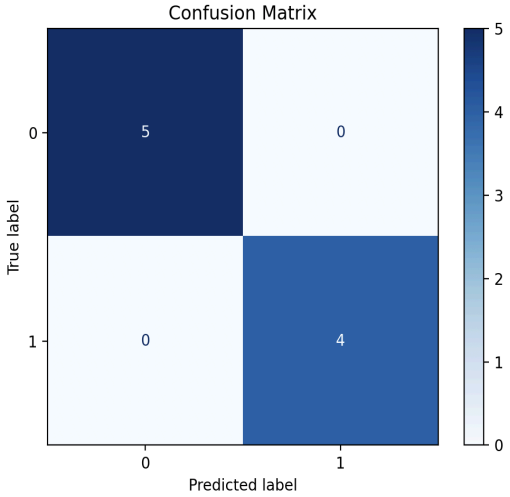


Figure 6: Confusion matrix for risk detection.

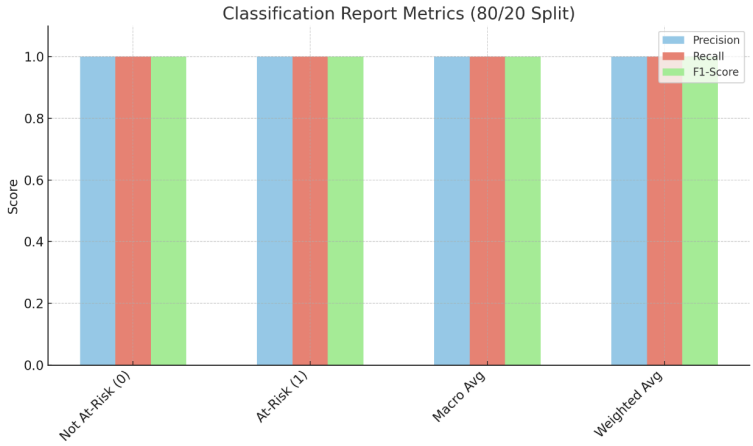


Figure 6: Classification report for risk detection containing the precision score, recall, and F1-scores.

To identify students potentially at emotional risk, we developed a risk detection module based on the outputs of the emotion classification model. Each student contributed five drawings from the random simulated data set, and any drawing predicted as a negative emotion (Angry, Fear, or Sad) was flagged as 1. If a student had three or more out of five drawings flagged as negative (≥60%), they were labeled as "At-Risk"; otherwise, they were labeled as "Not At-Risk." The average emotion indicators across the five drawings were used as input features for a Random Forest Classifier. Using an 80/20 train-test split, the model achieved 100% accuracy, precision, recall, and F1-score on the test set of 9 students (45 images). The confusion matrix showed perfect classification for both At-Risk and Not At-Risk groups, and the classification report confirmed perfect scores across all evaluation metrics.

5.7 Error Analysis

The risk detection module achieved 100% accuracy, precision, recall, and F1-score on the test set, with perfect classification of both At-Risk and Not At-Risk students. However, due to the small size of the test set (n = 9), the model's stability and generalizability remain uncertain. Such perfect results may not generalize well to larger, more heterogeneous populations. Future work should involve analyzing feature importance to understand which emotional patterns contribute most to risk classification. Additionally, evaluating the model on new, unseen data and conducting systematic hyperparameter tuning will be essential steps toward improving robustness and ensuring

reliable performance in practical deployment settings.

6. Discussion and Analysis

The developed model demonstrates significant capability in automated emotional assessment via drawings. However, performance disparities highlight the complexity of interpreting emotions visually. Future work should address limitations by expanding the dataset, including additional metadata such as age and location, and exploring temporal analysis of sequential drawings. The risk detection approach provides a practical screening tool that, while currently promising, necessitates validation on larger cohorts.

7. Conclusion

The developed model demonstrates notable potential for automated emotional assessment through children's drawings. Nonetheless, the observed variation in classification performance across different emotion categories underscores the inherent complexity of interpreting abstract visual expressions. These disparities suggest that certain emotions, such as fear, may lack distinct visual cues, posing challenges for feature-based learning.

Future improvements should focus on addressing current limitations by expanding the dataset size and diversity, incorporating relevant metadata such as age, gender, or geographic background for each drawing to provide contextual information, and investigating longitudinal patterns across sequential drawings to capture temporal emotional dynamics. Additionally, while the risk detection component yielded perfect performance on the test set, further validation on larger and more heterogeneous populations is essential to confirm its reliability and generalizability in real-world settings.

8. Contribution Statement

- **Sadia Fathima**: Resized and normalized images; generated charts; developed research inference logic; ran prediction scripts on test data; built and compiled risk detection model; trained and evaluated.
- **Jeslyn Liew**: Converted all file types to a unified format; divided data into training and testing sets; built, trained, and fine-tuned model; compiled model; integrated dropout layers; trained with callbacks; organized and led team meetings.
- **Yiming Zhang**: Encoded emotion labels; researched and evaluated algorithms; shared findings and recommended algorithmic approaches; created data augmentation pipeline; evaluated final model; visualized predictions and results.

9. References

- Savchenko, A. V. (2023). Facial Expression Recognition with Adaptive Frame Rate Based on Multiple Test Correction. PMLR. https://proceedings.mlr.press/v202/savchenko23a/savchenko23a.pdf
- Zhang, Q., Xu, C., Jin, J., Zhao, W., & Tang, J. (2024). Visual Prompting in LLMs for Enhancing Emotion Recognition. EMNLP. https://aclanthology.org/2024.emnlp-main.257.pdf
- Wen, J., Zhang, X., Li, M., & Wang, Z. (2023). Learning More from Mixed Emotions: A Label Refinement Method for Emotion Recognition in Conversations. TACL. https://aclanthology.org/2023.tacl-1.84/
- Perera, V. (2022). Children Drawing Dataset. Kaggle. https://www.kaggle.com/datasets/vishmiperera/children-drawings/data

10. GitHub Repository and Google Drive

GitHub:
https://github.com/xiaoyujes/ECS-111-Emotion-Detection
Google Drive:
https://drive.google.com/drive/folders/1xkOsVxCkwTQJi3ruOoOHVbvzSn8kqRCS?zx=dsjt3vfp82wq